

RFM Sales Customer Segmentation Report

Customer Recency, Frequency and Monetary Value Analysis using MySQL

Analysis Area	Details
Database	RFM_SALES
Source Table	SAMPLE_SALES_DATA
Analysis Period	06 January 2003 - 31 May 2005
Core Method	RFM scoring using NTILE(5)
Customer Dimensions	Recency, Frequency, Monetary Value
Output	Customer-level RFM scores and business segments

Purpose: This report documents the logic implemented in the supplied SQL query and explains how the resulting RFM customer segments can be used for sales and customer-management decisions.

1. Executive Summary

The SQL analysis builds a customer-level RFM model from the SAMPLE_SALES_DATA table. RFM stands for Recency, Frequency and Monetary Value and is a widely used framework for understanding customer engagement and purchasing value.

The analysis first determines the business-date range, then aggregates transaction data by customer. Each customer receives a Recency, Frequency and Monetary score from 1 to 5 using MySQL's NTILE window function. The three scores are combined into an RFM score and mapped to business-oriented customer segments such as Champions, Loyal Customers, Potential Loyalists, Promising Customers, Needs Attention, and About to Sleep.

The resulting RFM view can then be joined back to the sales table to compare customer segments by quantity ordered and sales amount. This creates a practical bridge between analytical segmentation and commercial performance.

Key dates identified by the SQL

Metric	Result
First business day in dataset	2003-01-06
Last business day in dataset	2005-05-31
Reference date for Recency	2005-05-31

2. RFM Methodology

2.1 Recency

Recency measures how recently a customer placed an order. The query uses the maximum order date for each customer and calculates the number of days between that date and the latest order date available in the dataset (31 May 2005). A lower Recency value means the customer purchased more recently.

2.2 Frequency

Frequency is calculated as the count of distinct ORDERNUMBER values for each customer. This measures how often the customer made separate purchases during the analysis period.

2.3 Monetary Value

Monetary Value is the total SALES amount generated by each customer. The SQL rounds the result to the nearest whole unit for reporting purposes.

2.4 RFM Scoring

Dimension	SQL ranking direction	Interpretation
Recency	NTILE(5) ORDER BY RECENCY_VALUE DESC	Higher score = more recent customer
Frequency	NTILE(5) ORDER BY FREQUENCY_VALUE ASC	Higher score = more frequent customer
Monetary	NTILE(5) ORDER BY MONETARY_VALUE ASC	Higher score = higher customer value

Important: The Recency ordering is intentionally reversed relative to Frequency and Monetary. Because a smaller number of days means greater recency, descending order assigns the higher R score to customers with fewer days since their last purchase.

3. SQL Workflow

1. Inspect the sales data and establish the first and last order dates.
2. Aggregate sales at customer level to calculate Recency, Frequency and Monetary Value.
3. Use NTILE(5) to divide customers into five approximately equal scoring groups.
4. Create the total RFM score by adding R, F and M scores.
5. Create a three-digit RFM combination such as 455 or 344.
6. Map RFM combinations to business customer segments using a CASE expression.
7. Store the final customer-level result in the RFM view.
8. Join RFM segments back to transaction data to evaluate quantity ordered and sales amount by segment.

4. Customer Segments

Segment	Business interpretation
Champions	Customers with very strong recent activity, purchase frequency and/or monetary value. They are hi
Loyal Customers	Customers showing sustained purchasing behavior and meaningful value. They are good candidate
Potential Loyalists	Customers with promising purchase behavior who may become loyal customers with the right enga
Promising Customers	Customers with moderate or developing value. Targeted offers and relevant product recommendati
Needs Attention	Customers whose activity or value suggests a risk of disengagement. Win-back or personalized cam
About to Sleep	Customers with weak RFM profiles and signs of inactivity. Low-cost reactivation campaigns should b
OTHER	RFM combinations not explicitly included in the supplied CASE statement. These customers require

5. Output and Business Analysis

The RFM view provides one record per customer containing the customer name, Recency Value, Frequency Value, Monetary Value, R/F/M scores, Total RFM Score, RFM Score Combination and Customer Segment.

The supplied SQL also produces two segment-level analyses. The first aggregates Monetary Value by customer segment and reports both total and average monetary value. The second joins the RFM view to the original sales data and summarizes total quantity ordered and total sales amount by segment.

Recommended management interpretation

- Champions: prioritize retention, premium service, early-access offers and referral/advocacy programs.
- Loyal Customers: strengthen loyalty through bundles, cross-selling and personalized product recommendations.
- Potential Loyalists: encourage repeat purchases with relevant follow-up campaigns and incentives.
- Promising Customers: improve conversion and basket size using targeted promotions and recommendations.
- Needs Attention: investigate declining activity and test win-back communications.
- About to Sleep: use low-cost reactivation campaigns and monitor response before increasing spend.
- OTHER: review the unclassified combinations to determine whether the segmentation rules should be expanded.

6. Data and Technical Considerations

- The Recency reference point is the maximum order date in the dataset rather than the current calendar date. This is appropriate for an historical dataset because it avoids artificially making every customer appear more inactive as time passes after the dataset ends.
- NTILE(5) creates relative scores based on the customer population. Therefore, an RFM score represents a customer's position relative to other customers in this dataset, not an absolute business threshold.
- The CASE statement explicitly lists selected three-digit RFM combinations. Any combination not listed becomes OTHER. This should be reviewed to ensure no strategically important customer group is unintentionally left unclassified.
- The final sales-by-segment query joins the customer-level RFM view to transaction-level sales. Since each customer has one RFM record, the join does not multiply transactions, assuming CUSTOMERNAME uniquely identifies the customer.
- Customer names are used as the join key. In a production environment, a stable customer ID would generally be safer because names can contain spelling variations or duplicates.
- The report does not invent segment totals because the supplied SQL was provided without the actual query result sets. The numerical segment totals should be taken directly from the executed RFM aggregation queries.

7. Conclusion

The SQL script establishes a complete RFM-based customer segmentation workflow in MySQL. It transforms transaction-level sales data into actionable customer groups by measuring how recently customers purchased, how frequently they purchased and how much revenue they generated.

The resulting RFM view can serve as a foundation for customer retention, loyalty management, campaign targeting and sales-performance reporting. The most important next step is to execute the final aggregation queries and incorporate the resulting segment-level totals, averages and sales/quantity measures into a management dashboard or periodic customer-performance report.

Prepared from the SQL logic supplied by the user. Analysis period: 06 January 2003 to 31 May 2005.